Topology
Home Work 6

Student: 26-712-224

October 22, 2026

"I wrote these solutions fully myself und would like to receive feedback on them"

Exercise 1

Consider the following subsets of $\mathbb{R}$, equipped with the Euclidean topology:

$$A = \{0\} \cup \left\{ \frac{1}{n} : n \in \mathbb{N},\ n \geq 1 \right\}, \qquad B = A \setminus \{0\}.$$

i) Prove directly from the definition of compactness using finite open subcovers that A is compact.

ii) Show that B is not compact by constructing an open cover that admits no finite subcover.

Answer:
(i) Let $\{U_i\}_{i \in I}$ be an open cover of A. Then we know that $0 \in U_{j_0}$ for some $j_0 \in I$. As U_{j_0} is open there exists an $\epsilon > 0$ such that $B(0, \epsilon) \subset U_{j_0}$.

Clearly, there is an $N \in \mathbb{N}$ such that $n > N \implies \frac{1}{n} < \epsilon$. This means that all terms from A after the Nth are in U_{j_0}. Obviously, there are at most N open sets from the cover needed to cover the first N elements of A.
I.e. we have found finite a subcover that covers A.

□

(ii) Consider the open cover for B: $\{U_i\}_{i \in \mathbb{N}}$ where $U_i = (\frac{1}{i}, 2)$. If we assume there is a finite subcover, let U_N be the highest index element in the cover.
Then the total cover can only be: $(\frac{1}{N}, 2)$ - and there will be elements from B that are not in this subcover.
Hence, this cover for B has no finite subcover.

□

Exercise 2

Consider $\mathbb{R}_\infty = \mathbb{R} \cup \{\infty\}$ with topology $\mathcal{U}_{\mathbb{R}_\infty}$ as defined in Problem Sheet 5, Exercise 1. Prove directly from the definition via finite open subcovers that $\mathbb{R}_\infty$ is compact.

Answer:

Exercise 3

Which ones of the following spaces are compact? For this exercise, you can use results from the theory as well as the exercises from last weeks problem sheet.

i) $\overline{B}(0,1) \setminus \{0\} = \{(x,y) \in \mathbb{R}^2 : 0 < \|(x,y)\| \leq 1\}$, with the subspace topology induced by the Euclidean topology on $\mathbb{R}^2$.

ii) $\{(t\cos t, t\sin t) \in \mathbb{R}^2 : 1 \leq t < \infty\}$, with the subspace topology induced by the Euclidean topology on $\mathbb{R}^2$.

iii) $\{(x,y,z) \in \mathbb{R}^3 : x^2 + y^2 = 1 \text{ and } |z| \leq 3\}$, with the subspace topology induced by the Euclidean topology on $\mathbb{R}^3$.

iv) The projective space $\mathbb{R}P^n$.

Answer:
(i) The set defined is a subset of $\mathbb{R}^2$ and so must be closed and bounded in order to be compact.
This set is bounded but NOT closed - and hence NOT compact.

(ii) This set is also a subset of $\mathbb{R}^2$ and is NOT bounded - and hence NOT compact.

(iii) Here, with the defined induced topology, we have both a bounded and closed set - it is the set for a bounded/closed cylindrical surface.
Hence, this set IS compact.

(iv) We know that the topological quotient map $S^n \to S^n/\{\pm 1\}$ - with equivalence $(x \sim -x)$ - is both continuous and surjective. As S^n is compact we must conclude that $S^n/\{\pm 1\}$ is also compact.
In last week's exercise sheet it was established that $\mathbb{R}P^n \cong S^n/\{\pm 1\}$ - i.e. the two spaces are homeomorphic. We must, therefore, conclude that $\mathbb{R}P^n$ is also compact.

Exercise 4

In this exercise we look at the compactness of some unusual topological spaces.

i) Equip a set X with the cofinite topology $\mathcal{U}_{cof}$. Show that X is compact, and show that each subset of X is compact. (Recall that $\mathcal{U}_{cof} = \{U \subseteq X : X \setminus U \text{ is finite}\} \cup \{\emptyset\}$.)

ii) Let now X be uncountable with the same cofinite topology $\mathcal{U}_{cof}$. Using the previous exercise, show that there are subsets which are compact but not closed.

Answer:
(i) Let $\{U_i\}_{i\in I}$ be an open cover of X. Take any one of these open non-empty sets - call it U_{j_0}. Then there are only finitely many elements in $X \setminus U_{j_0}$.
If there are $N < \infty$ elements not covered, then with at most N open sets from $\{U_i\}_{i\in I}$ we can create a finite subcover for X. Hence, X is compact.

Now let S be a **proper** subset of X. By intersecting S with each open set in X we find that it generates the "cofinite" topology on S. As in the case above we must therefore conclude that S is itself compact (using the induced topology).

(ii) Here X is uncountable. From (i) we know that X and any subset of X are compact. Now let $Y \subset X$ be open, then $Z = X \setminus Y$, by definition, is closed. But $Z \subset X$ and thus is compact.

Exercise 5

Let (X, d) be a metric space, $K \subset X$ compact, $F \subset X$ closed, and $K \cap F = \emptyset$. Show there is $\delta > 0$ with $d(x, y) \geq \delta$ for all $x \in K$, $y \in F$.

Answer:
We know that $d(k, f) > 0, \quad \forall k \in K; \forall f \in F$, since d is positive definite AND $d(k, f) = 0 \iff k = f$, which can never be the case as $K \cap F = \emptyset$.

It should be obvious from its definition that it must be continuous (in the $\delta - \epsilon$ definition) - here we would just take $\delta = \epsilon$.

Let us fix $f \in F$, then $D_f(k) = d(k, f)$ is a simple function: $D_f : K \to \mathbb{R}_{>0}$ which must also be continuous. Since K is compact there is a $k_f \in K$ for which this function attains a minumim value. Then $D_f(k_f) = d(k_f, f) = \delta_f > 0; \ \forall f \in F$.

In this was we have that the image of $K \times F$ is a subset of $\mathbb{R}_{>0}$ and hence must be bounded below. Consequently:

$$\delta = \text{Inf}_{f \in F}\{\delta_f\} \geq 0$$

For the sake of argument let us say that $\delta = 0$. Then it will be possible to find a sequence of points: $\{(k_n, f_n)\} \subset K \times F$ whose image approaches 0. Now $\{k_n\} \subset K$ and as K is compact it will have a subsequence* $\{k_{i_j}\}$ which converges to some $k \in K$. Then we have that:

$$d(k, f_{n_j}) \leq d(k, k_{n_j}) + d(k_{n_j}, f_{n_j}) \tag{5.1}$$

Now the two terms on the RHS of (5.1) both tend to zero - i.e. can be made arbitrarily small - and this can only hold if k is an accumulation point of F. But F is closed and hence we must have that $k \in F$.
Contradiction as $K \cap F = \emptyset$ and hence we must have $\delta > 0$.

□

* This is true in a metric space.

Exercise 6

Consider a topological space $(Y, \mathcal{U})$ and consider its infinite product $X = \prod_{i=1}^{\infty} Y$ equipped with the box topology $\mathcal{U}_{box}$ (see also Problem Sheet 5, Exercise 5).

i) Let $Y = \{0, 1\}$ with the discrete topology. Show that $(X, \mathcal{U}_{box})$ is not compact using finite open subcovers.

ii) Let $Y = [0, 1]$ with the standard euclidean topology. Show that $(X, \mathcal{U}_{box})$ is not compact.

Answer: